A KISSING CONFIGURATION OF 200540 POINTS IN $\mathbb{R}^{27}$

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Abstract. We show that the kissing number in 27 dimensions satisfies $K(27) \geq 200540$, improving the record of 200044 by 496. The record configurations in dimensions 25 through 31 all have a common form, and the number of points in such a configuration is governed by a single integer m. We prove that $m \leq \lfloor 2K(k)/3 \rfloor$ in dimension $24+k$, under hypotheses that all seven published configurations satisfy, and find that all seven attain this bound except in dimension 27, where m falls one short. Closing that gap needs no new coordinates, only a regrouping of the configuration already published, and it adds 496 points. For $k = 5, 6, 7$ the value of $K(k)$ is not known, and the comparison there is against the best lower bound known for it. Every assertion made here about the configurations is verified in exact arithmetic over $\mathbb{Z}[\sqrt{2}, \sqrt{3}, \sqrt{6}]$, the ring in which the published coordinates lie.

The kissing number $K(n)$ is the largest number of unit vectors in $\mathbb{R}^n$ whose pairwise inner products are all at most $\frac{1}{2}$. We prove the following, improving the record of 200044 held by the configuration of [6] by 496.

Theorem. *$K(27) \geq 200540$.*

This is Theorem 8 below. Section 1 describes the family of constructions the record configurations in dimensions 25 through 31 all belong to, Section 2 bounds the quantity that governs their size, Section 3 gives the configuration, and Section 4 verifies it.

1. The construction family

Write $\mathbb{R}^{24+k} = \mathbb{R}^{24} \oplus \mathbb{R}^k$ and split a unit vector as $p = (v, w)$. Let $X \subset S^{23}$ denote the 196560 minimal vectors of the Leech lattice Λ_{24}, normalised to unit length; recall that $\langle x, x' \rangle \in \{0, \pm\frac{1}{4}, \pm\frac{1}{2}, \pm 1\}$ for $x, x' \in X$ [3]; we re-verified this value set in exact arithmetic for the copy of X recovered from [1].

Those configurations all have the following form, which we extracted directly from the published coordinates [1] for every $k = 1, \dots, 7$:

- an *equator* $\{(x, 0) : x \in X \setminus D\}$, for a set $D \subset X$ of *deleted* vectors;

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- *caps* $\big(\sqrt{\tfrac{2}{3}}\,u,\ \sqrt{\tfrac{1}{3}}\,\widehat{w}\big)$, in which $u \in X$ is the cap's *head* and the unit vector $\widehat{w} \in \mathbb{R}^k$ its *direction*;
- *axis points* $(0, \omega)$, the ω forming a kissing configuration in $\mathbb{R}^k$: at most $K(k)$ of them, further constrained by (3) below.

Let $W \subset S^{k-1}$ be the set of directions that occur and, for $\widehat{w} \in W$, let $U_{\widehat{w}} \subset X$ be the set of heads occurring with $\widehat{w}$, the *class* of $\widehat{w}$. The constraints are as follows; all are immediate from $\langle p, p' \rangle \le \frac{1}{2}$.

Lemma 1. *The above configuration is a kissing configuration if and only if*

$$(1) \qquad (E) \qquad u \in D \text{ for every } u \in \textstyle\bigcup_{\widehat{w}} U_{\widehat{w}},$$

$$(2) \qquad (C) \qquad \tfrac{2}{3}\langle u, u' \rangle + \tfrac{1}{3}\langle \widehat{w}, \widehat{w}' \rangle \le \tfrac{1}{2} \quad \text{for caps } (u, \widehat{w}) \ne (u', \widehat{w}'),$$

$$(3) \qquad (A) \qquad \langle \omega, \widehat{w} \rangle \le \tfrac{\sqrt{3}}{2} \quad \text{for all axis points } \omega \text{ and } \widehat{w} \in W.$$

Proof. An equatorial point $(x, 0)$ and a cap $(\sqrt{2/3}\,u, \sqrt{1/3}\,\widehat{w})$ have inner product $\sqrt{2/3}\,\langle x, u \rangle$, which is at most $\sqrt{2/3} \cdot \frac{1}{2} < \frac{1}{2}$ when $x \ne u$, and equals $\sqrt{2/3} > \frac{1}{2}$ when $x = u$; this gives (1). Two caps give (2) directly, and an axis point and a cap give $\sqrt{1/3}\,\langle \omega, \widehat{w} \rangle \le \frac{1}{2}$, which is (3). Equatorial pairs and axis pairs are inherited from X and from the kissing configuration in $\mathbb{R}^k$, and $(x, 0) \perp (0, \omega)$. ☐

Condition (1) is the essential point: the head of a cap must be deleted from the equator. Only then may the cap sit at $|v| = \sqrt{2/3} > \frac{1}{2}$.

Remark 2. Condition (3) admits no axis point at all when $k = 1$: there $W = \{\pm 1\}$, so either choice of $\omega = \pm 1$ has $\langle \omega, \widehat{w} \rangle = 1 > \frac{\sqrt{3}}{2}$ for one of the two cap directions. This is why the $k = 1$ row of Table 1 carries no axis points where every other row carries $K(k)$ of them.

Specialising (2) to $\widehat{w}' = \widehat{w}$ gives $\langle u, u' \rangle \le \frac{1}{4}$: every class has pairwise cosines at most $\frac{1}{4}$. Write $\mathcal{S} = \{U_{\widehat{w}} : \widehat{w} \in W\}$ for the set of distinct classes.

Proposition 3. *If the classes in $\mathcal{S}$ are pairwise disjoint and all have size N, and $D = \bigcup_{\widehat{w}} U_{\widehat{w}}$, then the configuration has*

$$\begin{aligned}\big(196560 - |\mathcal{S}|\,N\big) \ + \ |W|\,N \ + \ (\#\ \textit{axis points}) \\ = \ 196560 + \big(|W| - |\mathcal{S}|\big)N + (\#\ \textit{axis points})\end{aligned}$$

points. In particular the net gain is governed by $m := |W| - |\mathcal{S}|$.

Each direction contributes N points, while each *distinct* class costs N deletions from the equator; a class reused on several directions is paid for only once.

2. The bound on m

Lemma 4. *Let $\widehat{w} \ne \widehat{w}'$ in W.*

(1) *If $U_{\widehat{w}} \cap U_{\widehat{w}'} \ne \emptyset$ then $\langle \widehat{w}, \widehat{w}' \rangle \le -\frac{1}{2}$.*

(2) *If there are* $u \in U_{\widehat{w}}$ *and* $u' \in U_{\widehat{w}'}$ *with* $\langle u, u'\rangle = \frac{1}{2}$, *then* $\langle \widehat{w}, \widehat{w}'\rangle \le \frac{1}{2}$.

Proof. (i) Taking $u = u'$ in the intersection in (2) gives $\frac{2}{3} + \frac{1}{3}\langle \widehat{w}, \widehat{w}'\rangle \le \frac{1}{2}$. (ii) Applying (2) to that pair gives $\frac{1}{3} + \frac{1}{3}\langle \widehat{w}, \widehat{w}'\rangle \le \frac{1}{2}$. □

In (ii) it is essential that u and u' lie in *different* classes: all cosines within a class are at most $\frac{1}{4}$, so a pair at cosine $\frac{1}{2}$ inside a single class does not exist, and the hypothesis read that way would be unsatisfiable.

Theorem 5. *Let a configuration be as in Proposition 3, so that the classes* $U_{\widehat{w}}$ *are pairwise equal or disjoint. Then*

(1) $m = |W| - |\mathcal{S}| \le |W| - \lceil |W|/3 \rceil = \lfloor 2|W|/3 \rfloor$;
(2) *if in addition any two* distinct *classes* $U_{\widehat{w}} \ne U_{\widehat{w}'}$ *contain a cross pair* $u \in U_{\widehat{w}}$, $u' \in U_{\widehat{w}'}$ *with* $\langle u, u'\rangle = \frac{1}{2}$, *then* $|W| \le K(k)$ *and hence* $m \le \lfloor 2K(k)/3 \rfloor$.

Proof. (i) Group the elements of W according to which class they carry; this is a partition because the classes are pairwise equal or disjoint. By Lemma 4(i) the members of a group are pairwise at cosine at most $-\frac{1}{2}$, so if a group has t members then $0 \le \left|\sum \widehat{w}\right|^2 \le t + 2\binom{t}{2}\left(-\frac{1}{2}\right) = t - \frac{t(t-1)}{2}$, whence $t \le 3$. (This uses no property of $\mathbb{R}^k$ beyond the inner product, so it holds in every dimension.) Therefore $|\mathcal{S}| \ge \lceil |W|/3 \rceil$.

(ii) Two directions carrying the same class are at cosine at most $-\frac{1}{2}$ by Lemma 4(i); two carrying different classes are at cosine at most $\frac{1}{2}$ by Lemma 4(ii). So W is a set of unit vectors in $\mathbb{R}^k$ that are pairwise at angle at least 60°, i.e. $|W| \le K(k)$. Since $n \mapsto \lfloor 2n/3 \rfloor$ is nondecreasing, (i) gives the stated bound. □

Remark 6. Neither hypothesis can be dropped. Without the first, the classes carried by the elements of W need not be equal or disjoint, so the grouping used in the proof of (i) is not a partition. Without the second, two disjoint classes all of whose cross cosines are at most 0 impose, through (2), only $\langle \widehat{w}, \widehat{w}'\rangle \le \frac{3}{2}$, which is no constraint; $|W|$ could then exceed $K(k)$. Both hypotheses do hold for all seven published configurations, as recorded in Table 1.

Remark 7. The kissing number is known exactly only in dimensions 1, 2, 3, 4, 8 and 24; that is what separates the two blocks of Table 1. In the lower block the entries 40, 72, 126 are the best known lower bounds for $K(5)$, $K(6)$, $K(7)$, each realised by the root system named in the same row, and the best known upper bounds are 44 and 134 [7] and 77 [4], as recorded in [2]. Since $n \mapsto \lfloor 2n/3 \rfloor$ is nondecreasing, the bound there is understated: were K to equal its upper bound, the entries would read 29, 51, 89 and m would fall short by 3, 3, 5. So in that block "m meets the bound" asserts only that it meets the value computed from the lower end of the range.

The distinction is not bookkeeping. What Theorem 5(i) constrains is $|\mathcal{S}| \ge \lceil |W|/3 \rceil$, and six of the seven rows already sit at that minimum:

k	$24+k$	W	$\lvert W\rvert$	$\lvert\mathcal{S}\rvert$	m	$\lfloor 2K(k)/3\rfloor$	$K(k)$	points
$K(k)$ known exactly								
1	25	A_1	2	1	1	1	2	197056
2	26	A_2	6	2	4	4	6	198550
3	27	A_3	12	5	**7**	**8**	12	200044
4	28	D_4	24	8	16	16	24	204520
$K(k)$ known only within the range shown								
5	29	D_5	40	14	26	26–29	40–44	209496
6	30	E_6	72	24	48	48–51	72–77	220440
7	31	E_7	126	42	84	84–89	126–134	238350

TABLE 1. The bound of Theorem 5(ii) against the values of m in the record configurations of [6, 1], all seven computed from the published coordinates in exact arithmetic. The two middle columns are the comparison, and they agree in every row but $k=3$, where $m=7$ falls one short of 8 — the slack this note removes. In every row W is closed under negation and has the cosine distribution of the roots of the named root system, so $|W|$ equals $K(k)$ in the upper block and the best known lower bound for $K(k)$ in the lower one, where the comparison is accordingly conditional on the configurations named in the W column. Throughout, the classes are pairwise disjoint with $N=496$; the axis points number $K(k)$ for $k\geq 2$ and none for $k=1$ (Remark 2); and both hypotheses of Theorem 5 hold, the maximum cross cosine over all 1267 pairs of distinct classes arising at $k=1,\dots,7$ being exactly $\frac{1}{2}$.

$|\mathcal{S}| = 1, 2, \mathbf{5}, 8, 14, 24, 42$ against $\lceil |W|/3\rceil = 1, 2, \mathbf{4}, 8, 14, 24, 42$. Only $k=3$ spends a class it need not, and that is the whole of the improvement below. For $k=5,6,7$ a larger m therefore cannot come from regrouping a given W; it would need a larger W, that is, a kissing configuration in $\mathbb{R}^k$ beating D_5, E_6 or E_7 — the open problem that put those rows in the lower block to begin with. Rows $k=1,2,3,4$ use the proven values $K=2,6,12,24$, so the slack at $k=3$ is unconditional.

3. DIMENSION 27

We extracted the published 200044-point configuration in $\mathbb{R}^{27}$ from [1] and computed its structure. It consists of 194080 equatorial points, 5952 caps with $|w| = 1/\sqrt{3}$ exactly, and 12 axis points. The caps lie on $|W| = 12$ directions whose pairwise cosines lie in $\{0, \pm\frac{1}{2}, \pm 1\}$, so W is a cuboctahedron — the A_3 root system of Table 1 — and $|W| = K(3) = 12$ is maximal. Each class has $N = 496$ elements and there are $|\mathcal{S}| = 5$ of them, pairwise disjoint, whose $5\cdot 496 = 2480$ vectors are exactly the $196560 - 194080 = 2480$ deleted from the equator. Thus $m = 12 - 5 = 7$, in agreement with Proposition 3: $196560 + 7 \cdot 496 + 12 = 200044$.

The grouping of the twelve directions is $\{3, 3, 2, 2, 2\}$, with pairwise cosine $-\frac{1}{2}$ inside each triple and -1 inside each pair. By the proof of Theorem 5(i) a group has at most three members, so the minimum possible number of groups is $\lceil 12/3 \rceil = 4$. We verified by exhaustive search over the $\binom{12}{3}$ triples and $\binom{12}{4}$ quadruples that the twelve directions contain exactly eight triples that are pairwise at cosine $-\frac{1}{2}$ (and no set of four with all cosines $\leq -\frac{1}{2}$) and that these eight triples admit exactly two partitions of W into four of them, namely

$$P_1: \quad \{\widehat{w}_0, \widehat{w}_7, \widehat{w}_{10}\}, \quad \{\widehat{w}_1, \widehat{w}_6, \widehat{w}_9\}, \quad \{\widehat{w}_2, \widehat{w}_3, \widehat{w}_{11}\}, \quad \{\widehat{w}_4, \widehat{w}_5, \widehat{w}_8\};$$

$$P_2: \quad \{\widehat{w}_0, \widehat{w}_8, \widehat{w}_9\}, \quad \{\widehat{w}_1, \widehat{w}_4, \widehat{w}_{11}\}, \quad \{\widehat{w}_2, \widehat{w}_5, \widehat{w}_{10}\}, \quad \{\widehat{w}_3, \widehat{w}_6, \widehat{w}_7\}.$$

Here $\widehat{w}_0, \ldots, \widehat{w}_{11}$ are the twelve directions in lexicographic order of their coordinate triples; the statement is of course independent of the labelling. Each of P_1 and P_2 reuses one of the two triples already present in the published grouping. Assigning any four of the five existing classes to the four triples of either partition yields $|\mathcal{S}| = 4$, and hence $m = 8$.

Theorem 8. $K(27) \geq 200540$.

Proof. Let $S_1, \ldots, S_4$ be any four of the five classes. Take $D = S_1 \cup S_2 \cup S_3 \cup S_4$, of size $4 \cdot 496 = 1984$; the equator $X \setminus D$ has 194576 points. Place S_i on the i-th triple of P_1 (or of P_2; the argument is identical), giving $12 \cdot 496 = 5952$ caps, and retain the 12 axis points. Equatorial pairs are inherited from X and axis pairs from the cuboctahedron of axis points; $(x, 0) \perp (0, \omega)$. Condition (1) holds by construction, since every cap head lies in D and the equator is $X \setminus D$. Condition (3) holds because W and the axis points are unchanged from the published configuration: $\max_{\omega, \widehat{w}} \langle \omega, \widehat{w} \rangle = \frac{1}{2} + \frac{\sqrt{2}}{4} = 0.8535\ldots < \frac{\sqrt{3}}{2} = 0.8660\ldots$, equivalently the largest inner product between a cap and an axis point is $\frac{\sqrt{3}}{6} + \frac{\sqrt{6}}{12} = 0.4927\ldots < \frac{1}{2}$. For (2) there are three cases, using that distinct elements of X have $\langle u, u' \rangle \leq \frac{1}{2}$ and that the four classes are pairwise disjoint, so that $u = u'$ can occur only within one class:

- $\widehat{w} = \widehat{w}'$: then $u \neq u'$ lie in one class, so $\langle u, u' \rangle \leq \frac{1}{4}$ and $\frac{2}{3} \cdot \frac{1}{4} + \frac{1}{3} = \frac{1}{2}$;
- $\widehat{w} \neq \widehat{w}'$ in the same triple: $\langle \widehat{w}, \widehat{w}' \rangle = -\frac{1}{2}$ and $\langle u, u' \rangle \leq 1$, giving $\frac{2}{3} - \frac{1}{6} = \frac{1}{2}$;
- $\widehat{w}, \widehat{w}'$ in different triples: they carry different classes, which are disjoint, so $u \neq u'$ and $\langle u, u' \rangle \leq \frac{1}{2}$; and $\langle \widehat{w}, \widehat{w}' \rangle \leq \frac{1}{2}$ for any two distinct elements of a cuboctahedron. Hence $\frac{2}{3} \cdot \frac{1}{2} + \frac{1}{3} \cdot \frac{1}{2} = \frac{1}{2}$, with equality exactly when both bounds are met.

The last case covers the remaining cuboctahedron cosines $0, -\frac{1}{2}, -1$ across triples as well, with strictly smaller values. Finally, an equatorial point and a cap have inner product $\sqrt{2/3}\,\langle x, u \rangle \leq \sqrt{2/3} \cdot \frac{1}{2} = 0.408\ldots$, since $x \neq u$ by (1). The total is $194576 + 5952 + 12 = 200540$. □

By Theorem 5 this value of m is optimal within the family: $|W| \le K(3) = 12$ and each group has at most three members, whence $m \le \lfloor 2 \cdot 12/3 \rfloor = 8$. Since $K(3) = 12$ is a theorem [8, 3], this optimality statement is unconditional, in contrast to the rows $k = 5, 6, 7$ of Table 1. The configuration just built satisfies the hypotheses of Theorem 5 as well as the published one does: its four classes are four of the five of [6], so they are still pairwise disjoint of size 496, and by Remark 6 any two of them contain a cross pair at cosine $\frac{1}{2}$.

Remark 9. Theorem 5 bounds m, not the number of points, which by Proposition 3 is $196560 + mN + (\#\text{ axis points})$ and so is linear in N. We prove nothing about N: we do not know the largest N for which X contains $\lceil |W|/3 \rceil$ pairwise disjoint N-element subsets of pairwise cosine at most $\frac{1}{4}$, and indeed we do not bound the size of even one such subset. All seven published configurations use $N = 496$ and the improvement here keeps it, so Table 1 compares configurations with the same N; a larger admissible N would give more points by the same counting. Finding one, or ruling it out, is the natural next question for this family.

4. Verification

The configuration is available as `data/dimension27_200540.txt` in the accompanying verification package [5], written in the format of [1]. The package proves the theorem twice, by different routes, with no floating-point arithmetic and no tolerance at any step.

`scripts/verify_configuration.py` works from the coordinate file alone, needs no third-party packages, and takes some fifteen seconds on a laptop; it is described below. `scripts/verify_exhaustive.py` assumes nothing about the geometry and compares all $\binom{200540}{2} = 20\,108\,045\,530$ pairs directly. Both also certify the published 200044-point configuration of [6] unchanged, which is a check on the verifiers as much as on the configuration.

Every coordinate in the dimension-27 block of [1] lies in $\mathbb{Z}[\sqrt{2}, \sqrt{3}, \sqrt{6}]$; the block uses 45 distinct coordinate tokens, all of that form. We represent the ring by integer 4-tuples in the basis $(1, \sqrt{2}, \sqrt{3}, \sqrt{6})$, so ring arithmetic is integer arithmetic; the comparisons that integers alone do not settle are settled by certified rational enclosures of the radicals.

Proposition 10. *After an exact rational rescaling, every point of the block in dimension* 27 *takes one of three forms*

$$\frac{(y, 0)}{\sqrt{32}}, \qquad \frac{(3y, t)}{\sqrt{432}}, \qquad \frac{(0, a)}{\sqrt{48}},$$

for the equatorial, cap and axis points respectively, where in the first two $y \in \mathbb{Z}^{24}$ with $y \cdot y = 32$, and where t and a lie in the ring with $t \cdot t = 144$ and $a \cdot a = 48$. Consequently each pairwise inner product is $\le \frac{1}{2}$ precisely when *the corresponding condition below holds:*

$$\begin{array}{llcl}
\textit{equator–equator} & \dfrac{y\cdot y'}{32} & \le \frac12 & \iff\ y\cdot y' \le 16,\\
\textit{equator–cap} & \dfrac{y\cdot y'}{16\sqrt6} & \le \frac12 & \iff\ y\cdot y' \le \lfloor 8\sqrt6\rfloor = 19,\\
\textit{cap–cap} & \dfrac{9\,y\cdot y' + t\cdot t'}{432} & \le \frac12 & \iff\ 9\,y\cdot y' \le 216 - t\cdot t',\\
\textit{equator–axis} & 0 & & \textit{always},\\
\textit{cap–axis} & \dfrac{t\cdot a}{144} & \le \frac12 & \iff\ t\cdot a \le 72,\\
\textit{axis–axis} & \dfrac{a\cdot a'}{48} & \le \frac12 & \iff\ a\cdot a' \le 24.
\end{array}$$

The equator–equator, equator–cap and cap–cap conditions are comparisons of integers; the cap–axis and axis–axis conditions are 12×12 *tables of comparisons in the ring.*

Proof. The classification into the three forms is exact and scale invariant: the equatorial points are the rows with $w = 0$, the axis points those with $v = 0$, and the caps those with $|w|^2/|p|^2 = \frac13$ as an identity between integers. Only then is a row rescaled, to squared norm 32, 432 or 48 according to its family. The block contains rows of nine distinct squared norms, namely 2, 3, 8, 12, 27, 32, 48, 108 and 432; the factors required are $4, 2, 1$ for the equatorial rows of norm $2, 8, 32$, then $12, 6, 4, 3, 2, 1$ for the cap rows of norm $3, 12, 27, 48, 108, 432$, and $4, 2, 1$ for the axis rows of norm $3, 12, 48$. Every factor is a positive integer, so the rescaling stays inside the ring, and a rescaled cap head is divisible by 3. The six displayed equivalences are then immediate on clearing denominators, using $\sqrt{32\cdot 432} = 48\sqrt6$ in the second, $\sqrt{432\cdot 48} = 144$ in the fifth, and $19^2 = 361 \le 384 < 400 = 20^2$ to evaluate $\lfloor 8\sqrt6\rfloor$. □

Remark 11. The scaling above is the only one used anywhere in this note or in the verification package. Fixing a single convention matters: an earlier draft of the package inherited the cap–axis threshold from a normalisation in which axis rows had squared norm 432, and so tested $t\cdot a \le 216$ rather than $t\cdot a \le 72$. Since $|t\cdot a| \le \sqrt{144\cdot 48} = 48\sqrt3 < 216$ always, that test could not fail, and the tightest condition in the configuration – $t\cdot a = 24\sqrt3 + 12\sqrt6 = 70.96\ldots$ against a limit of 72 – went unchecked. The package now derives each threshold from the squared norms of the two families involved, and also asserts of each that Cauchy–Schwarz permits a value above it, so that a vacuous test is itself a failure.

Remark 12. Where a script evaluates the integer dot products with a floating point matrix product, it is exact. The rational coordinates are bounded by 12, so each of the 24 products has absolute value at most 144 and every partial sum is an integer of absolute value at most 3456; integers of that size are represented exactly in IEEE 754 binary64, so no rounding occurs at any step, under any summation order and with or without fused multiply–add. The scripts do not assume this: each checks a sample of its products against exact integer arithmetic, and derives the bound from the data rather than assuming it.

The outcome is that all 200540 points are distinct and of norm exactly 1, and that every one of the 20 108 045 530 pairs satisfies $\langle p, q\rangle \le \frac{1}{2}$.

The fast script avoids the quadratic sweep by certifying, from the coordinates alone, that the 196560 heads are the minimal vectors of a copy of Λ_{24}. Call the multiset of absolute values of the nonzero coordinates of a vector its *shape*. The census by shape is 1104 of shape $(\pm 4^2)$, 97152 of shape $(\pm 2^8)$ and 98304 of shape $(\mp 3, \pm 1^{23})$, which is the census of Λ_{24}; the supports of those of shape $(\pm 2^8)$ are 759 octads, spanning a $[24, 12]$ binary code with the weight enumerator $1 + 759x^8 + 2576x^{12} + 759x^{16} + x^{24}$ of the extended Golay code and equal to its weight-8 words; and every head satisfies the congruences defining Λ_{24} with respect to that code. Two distinct minimal vectors then satisfy $y \cdot y' \le 16$, which disposes of the equatorial pairs. As a further check on the input, every one of the 196560 was confirmed to have the inner product distribution of the Leech minimal vectors: the numbers of x' with $\langle x, x'\rangle$ equal to $-1, -\frac{1}{2}, -\frac{1}{4}, 0, \frac{1}{4}, \frac{1}{2}, 1$ are $1, 4600, 47104, 93150, 47104, 4600, 1$, summing to 196560.

Equality is attained, and attained exactly. The pairs realising $\frac{1}{2}$ are the equatorial pairs at Leech cosine $\frac{1}{2}$, the 48 ordered axis pairs at cosine $\frac{1}{2}$, and three families of cap pairs:

$$\begin{array}{llll} \langle \widehat{w}, \widehat{w}'\rangle = 1, & \langle u, u'\rangle = \frac{1}{4}: & \frac{2}{3}\cdot\frac{1}{4} + \frac{1}{3}\cdot 1 = \frac{1}{2}, & 900096, \\ \langle \widehat{w}, \widehat{w}'\rangle = \frac{1}{2}, & \langle u, u'\rangle = \frac{1}{2}: & \frac{2}{3}\cdot\frac{1}{2} + \frac{1}{3}\cdot \frac{1}{2} = \frac{1}{2}, & 277408, \\ \langle \widehat{w}, \widehat{w}'\rangle = -\frac{1}{2}, & u = u': & \frac{2}{3}\cdot 1 + \frac{1}{3}\cdot\left(-\frac{1}{2}\right) = \frac{1}{2}, & 11904, \end{array}$$

where the counts are of ordered pairs. The third family is the one created by the regrouping: it consists of the $4 \cdot 6 \cdot 496$ pairs of caps that share a head and sit on two distinct directions of a single triple. Its presence is the reason a triple, and no larger set, is the most that can share a class.

Remark 13. The classes $S_1, \ldots, S_4$ are taken verbatim from the configuration of [6]: every Leech vector, direction and axis point used here already occurs there, and no coordinate was computed anew. The improvement is purely the observation that partitioning the twelve directions into four triples, rather than two triples and three antipodal pairs, makes the fifth class unnecessary and returns the 496 vectors it was deleting to the equator.

Use of AI assistance

The construction of Section 3 was found by Claude Opus 5 (Max effort), at the fortieth prompt of a session in which the author supplied the problem, Cohn's table [2] and data set [1]; most of the thirty-nine earlier prompts were some form of "try harder". The model maintained throughout that it would not succeed: "I have not produced a new SOTA … continuing would be me generating variations faster than either of us can check them." This note and the verification of Section 4 were then written by the author with Claude Code (Opus 5, Extra effort) and attacked by further instances of the same model as adversarial reviewers.

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